Bachelor of Computer Applications (Hons. Game Development)
Semester- III

Subject: Database Management System

Assignment 6: Implementation of Data Integrity Constraints

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1. Purpose of data integrity

Constraints prevent invalid or contradictory data from being stored. They make the database enforce important rules rather than relying only on users or application code.

Constraint (Based on purpose of data integrity)	Where used	Purpose
PRIMARY KEY	All tables; composite key in Book_Author	Uniquely identifies each record and prevents duplicate or NULL key values.
FOREIGN KEY	Books, BookCopies, Book_Author, Loans, Reservations, Fines	Ensures that referenced records exist in the related tables and maintains referential integrity.
UNIQUE	Members.email, Staff.email, Publishers.name, Categories.category_name, Books.isbn, Fines.loan_id	Prevents duplicate values in columns that must contain unique information.
CHECK	Membership status, copy status, reservation status, fine amount, publication year and loan dates	Ensures that only valid values and conditions are entered into the database.
DEFAULT	Members.membership_status, Books.language, BookCopies.status, Reservations.status, Fines.amount	Automatically assigns predefined values when no value is provided during data insertion.

2. SQL queries to verify constraints

```
-- A. Primary key: duplicate member_id should fail --
INSERT INTO Members (member_id, name, email, membership_date, membership_status)
VALUES (1, 'Duplicate Member', 'duplicate@example.com', '2026-10-20', 'Active');

-- B. UNIQUE: duplicate email should fail --
INSERT INTO Members (member_id, name, email, membership_date, membership_status)
VALUES (99, 'Email Duplicate', 'aarav.sharma@example.com', '2026-10-20', 'Active');

-- C. CHECK: invalid membership status should fail --
INSERT INTO Members (member_id, name, email, membership_date, membership_status)
VALUES (99, 'Invalid Status', 'invalid@example.com', '2026-10-20', 'Paused');

-- D. FOREIGN KEY: nonexistent category should fail --
INSERT INTO Books (book_id, title, isbn, publisher_id, category_id, publication_year, language)
VALUES (99, 'Unknown Category Book', '9999999999999', 1, 999, 2020, 'English');

-- E. CHECK: negative fine amount should fail --
INSERT INTO Fines (fine_id, loan_id, amount, reason, paid_status, payment_details)
```

```
VALUES (3999, 1001, -10.00, 'Late return', 'Unpaid', NULL);

-- F. UNIQUE: second fine for same loan should fail --
INSERT INTO Fines (fine_id, loan_id, amount, reason, paid_status, payment_details)
VALUES (3998, 1003, 50.00, 'Damage', 'Unpaid', NULL);

-- G. Verify foreign-key links with a query --
SELECT l.loan_id, m.name, bc.copy_id, b.title, s.name AS staff_name
FROM Loans l
JOIN Members m ON m.member_id = l.member_id
JOIN BookCopies bc ON bc.copy_id = l.copy_id
JOIN Books b ON b.book_id = bc.book_id
JOIN Staff s ON s.staff_id = l.staff_id
ORDER BY l.loan_id;



-- H. Verify defaults (membership_status omitted -&gt; Active) --
INSERT INTO Members (member_id, name, email, membership_date)
VALUES (90, 'Default Test', 'default.test@example.com', '2026-10-20');
SELECT member_id, membership_status FROM Members WHERE member_id = 90;
DELETE FROM Members WHERE member_id = 90;
```

3. Verification results

Tests A–F fail with duplicate-key, unique-key, check-constraint, or foreign-key errors. The join in G returns the six sample loan records with valid related rows. Test H inserts the row with membership_status = 'Active' by default, then deletes the temporary row.